
William Garzon

Information Technology — 10th semester

Repository https://github.com/Rechashhh/Workshop1_ProgramacionWeb

Live site https://rechashhh.github.io/Workshop1_ProgramacionWeb/

This report walks through the site section by section. For every change it states what the page looked like before, why the change was made, the CSS rule that produces it, and what else in the site that rule reaches.

1. What is being delivered

The repository holds two independent folders, each with its own README and its own index. Neither depends on the other, so each workshop can be opened and graded on its own.

```
Workshop1_ProgramacionWeb/
|
+-- HTML/                      WORKSHOP 1 - only HTML
|   +-- README.md              description of the workshop 1 content
|   +-- index.html
|   +-- pages/                  courses, schedule, contact, activities, map
|   +-- assets/                 images
|
+-- HTML+CSS/                   WORKSHOP 2 - HTML + CSS
|   +-- README.md              description of the workshop 2 content
|   +-- index.html
|   +-- courses.html schedule.html activities.html map.html contact.html
|   +-- css/style.css           the single stylesheet
|   +-- assets/                 images and the self-hosted typeface
|
+-- index.html                  entry page: links to both folders
```

The two folders hold the **same six pages**. Workshop 1 is the structure written in HTML5 alone; Workshop 2 is that same structure with a stylesheet on top. Reading them side by side is reading a before and after of identical markup.

Two workshops, one site

WILLIAM GARZON · INFORMATION TECHNOLOGY · SEMESTER II 2026

The same six-page website, delivered twice. Workshop 1 is the structure written in HTML5 alone. Workshop 2 is that same structure with a stylesheet on top, so the two folders read as a before and after of the very same markup.

Open a workshop

Workshop 1 – only HTML

Six interconnected pages with no CSS and no JavaScript. The semantic elements, the schedule table with its `rowspan` and `colspan`, and the contact form with its ten input types.

HTML5

Workshop 2 – HTML + CSS

The same markup with a stylesheet: design tokens, a light and a dark theme, an image viewer and a welcome sequence, all without a single line of JavaScript.

HTML5 · CSS

Repository

- github.com/Rechashhh/Workshop1_ProgramacionWeb (opens in a new tab)
- Each folder carries its own README describing what the workshop contains.

Figure 1. The entry page at the repository root. It exists so that the published site opens on a page that leads to both workshops.

Published with GitHub Pages

The site is live at https://rechashhh.github.io/Workshop1_ProgramacionWeb/

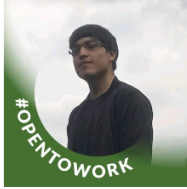
Pages serves from the repository root, and the home page of each workshop lives inside its folder, so the root `index.html` is what makes the link land somewhere useful. A `.nojekyll` file stops Pages from running the files through Jekyll: these are hand-written HTML pages and must be served exactly as they are.

Workshop 1, unchanged

The HTML folder is the Workshop 1 submission as it was delivered, taken from the commit tagged `workshop-1` in the repository. It has no stylesheet, which is what that workshop required.

My University Website

- [Home](#)
- [Courses](#)
- [Schedule](#)
- [Activities](#)
- [Map](#)
- [Contact](#)



William Garzon – Yachay Tech, 2026

William Garzon

Information Technology student

10 Semestre

About Me

Hi! I'm Willy Wonka, an IT student who needs money to spend on videogames, dance and chaulafán. I currently work as a Full Stack Developer at Ripconci, a national construction company. These days I'm focused on finishing my degree and learning more about web development fundamentals: HTML, CSS and JavaScript.

My personality type is ENFJ and my zodiac sign is Pisces.

Academic Interests

These are the topics I want to go deeper into during my last semester:

- Web development fundamentals: HTML, CSS and JavaScript
- Databases and data modeling
- Frontend frameworks and libraries

Figure 2. Workshop 1: the same content with no CSS. Every page reads top to bottom in order, the tables are still tables and the form still works.

2. The stylesheet

One file, `HTML+CSS/css/style.css`, around 1200 lines, organised in sections: reset, typography, design tokens, dark theme, base elements, the welcome screens, header and navigation, layout, lists, figures, cards, tables, form, image viewer, footer, responsive rules and reduced motion.

Design tokens

Every colour, measure and duration is declared once and referenced by name. This is the decision the whole stylesheet rests on.

`css/style.css`

```
:root {
  --bg:          #f5f6fa;      /* page background      */
  --bg2:         #ffffff;      /* panels                */
  --bg3:         #eef0f6;      /* recessed cells        */
  --border:      #e1e4ec;
  --border-hi:   #c7ccdc;
  --text:        #14161f;
  --text-dim:    #5b6072;
  --accent:      #a4161a;      /* the school's red      */
  --accent-bg:   rgba(164, 22, 26, 0.08);
  --max-width:   1280px;
  --measure:     68ch;         /* readable line length  */
  --font-body:   'Manrope', 'Segoe UI', system-ui, sans-serif;
  --font-px:     'Courier New', 'Consolas', monospace;
}
```

What it affects: changing `--accent` on that one line repaints headings, links, table headers, buttons, form focus rings, list markers and the caret in the welcome screen. Without tokens the same change would mean finding a literal colour in dozens of rules, and missing some.

Two themes, tokens only

`css/style.css`

```
@media (prefers-color-scheme: dark) {
  :root {
    --bg:    #0a0c0f;  --bg2: #111318;  --bg3: #191c23;
    --text:  #e8eaf0;  --text-dim: #9aa1b5;
    --accent: #f87171;
  }
}
```

The dark theme redefines **only the tokens**. Not one component rule is duplicated, which is why the two themes cannot drift apart: there is a single set of rules and two sets of values.

Self-hosted typeface

`css/style.css`

```
@font-face {
  font-family: 'Manrope';
  src: url('../assets/fonts/Manrope.ttf') format('truetype');
  font-weight: 200 800;
  font-display: swap;
}
```

Manrope ships inside the repository instead of being pulled from a CDN, so the site renders identically with no internet connection. If the file were missing the browser falls back to the list in `--font-body` and the page still holds its shape.

3. Section by section

The welcome sequence `index.html`

On arrival the visitor meets a single large PRINT button. Pressing it shows a greeting that types itself out, and after a few seconds the screen fades on its own and leaves the site at its home section.



Figure 3. *First screen: the button on its own.*



Figure 4. *After pressing it: the greeting types itself out, one character at a time.*

Why: the assignment for Workshop 1 was a “Hola, Mundo” exercise. Turning that into the door of the site keeps the reference and gives the page an opening moment.

How, with no JavaScript: two CSS mechanisms do all the work.

css/style.css

```
.intro-hello { display: none; }

/* :target is the one CSS selector that reacts to a click: it matches
   the element whose id is in the address bar. */
.intro-hello:target {
  display: flex;
  position: fixed; inset: 0; z-index: 100;
  animation: greet-exit 4.2s ease both;
}

@keyframes greet-exit {
  0%, 80% { opacity: 1; visibility: visible; }
  100%   { opacity: 0; visibility: hidden; }
}

/* With the greeting active, the button screen retires. "~" only looks
   forward, which is why #hello comes first in the HTML. */
.intro-hello:target ~ .intro-start { display: none; }

/* The typing effect animates WIDTH in discrete steps, one per
   character. It only works in a monospaced face, because the `ch`
   unit assumes every character is the same width.
   "Hola, Mundo!" = 12 characters -> 12ch and steps(12). */
.typewriter {
  font-family: var(--font-px);
  width: 12ch; overflow: hidden; white-space: nowrap;
  border-right: 3px solid var(--accent);
  animation: type 2.4s steps(12) 0.4s both,
             caret 0.8s step-end infinite;
}
```

What it affects: the sequence appears only on a first visit. Any active `:target` retires it — the menu's Home entry points to `index.html#home` precisely so that returning to the home page does not replay it.

Home index.html

Before: the page opened with a photograph and three loose lines (name, programme, semester). It did not say what the site was for until the visitor scrolled.

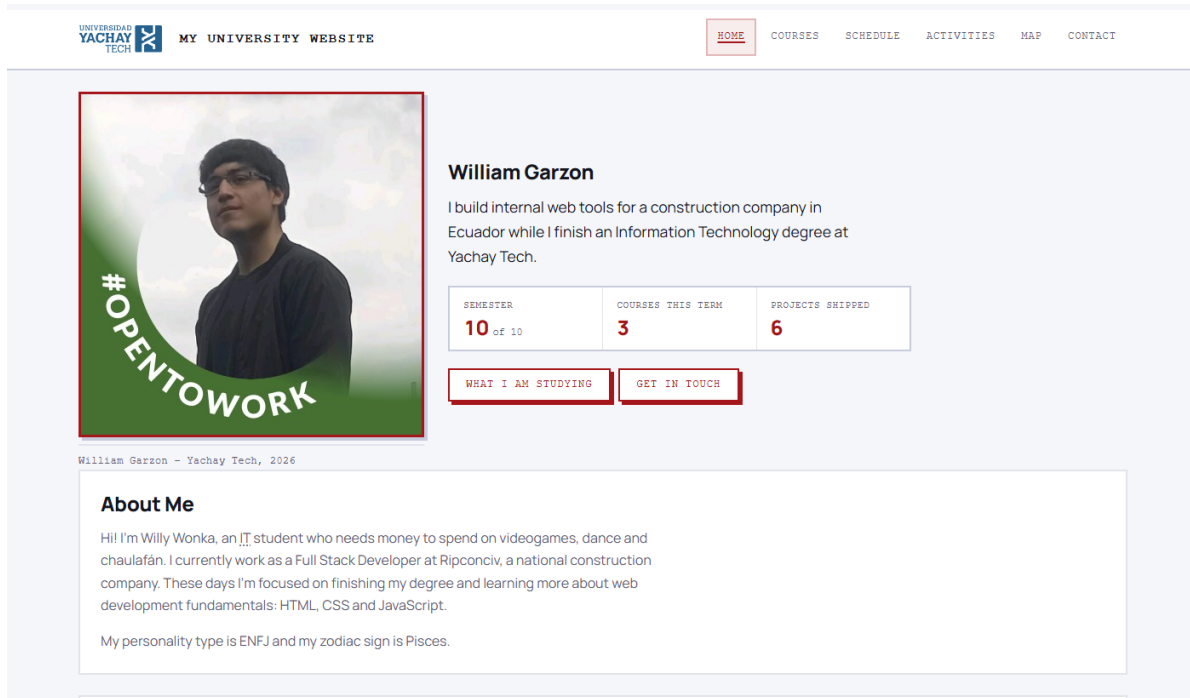


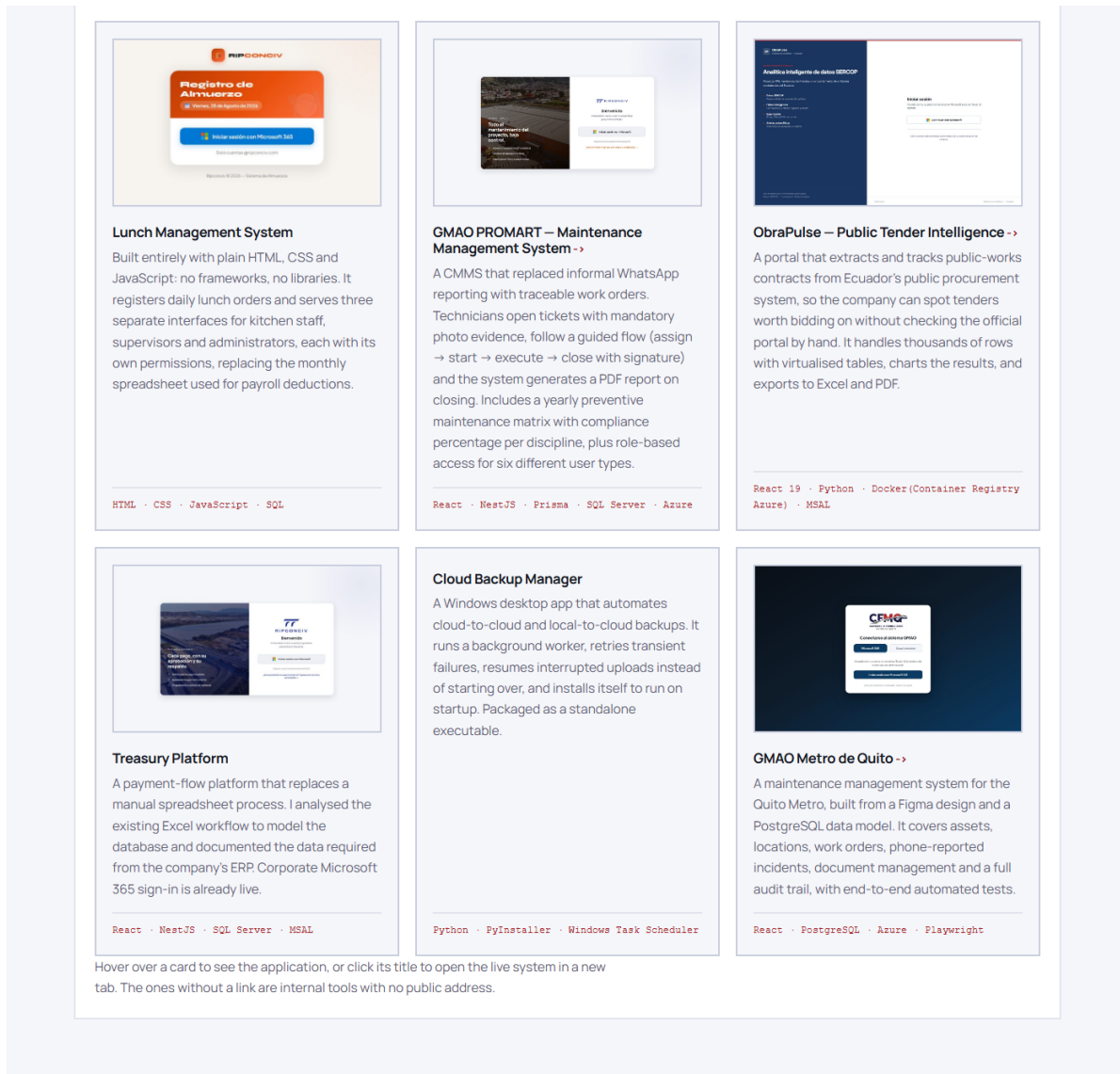
Figure 5. Home: a positioning line, three figures and two clear next steps.

Why: the first screen has one job, which is to answer “what is this and what can I do here”. A number with a small label answers in a second what a paragraph takes ten to say.

css/style.css

```
.stats {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(9rem, 1fr));
  gap: 1px;
  background: var(--border); /* the gap IS the divider line */
  border: 2px solid var(--border-hi); /* it shows through the gap */
}
.stat { background: var(--bg2); padding: 0.8rem 1rem; }
.stat dd {
  font-size: 1.5rem; font-weight: 800; color: var(--accent);
  /* tabular figures: numbers in different cards line up instead of
    shifting according to the digit */
  font-variant-numeric: tabular-nums;
}
```

What it affects: the same `.stats` block is reused on the schedule page for its five figures. One rule, two pages, no duplication.



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Web Applications - Workshop 2 - Hecho con HTML5 y CSS

Figure 6. The projects grid. Each thumbnail is a screenshot of the running system, and four titles link to the live application.

Before: the screenshot was a layer that only appeared on hover and covered the whole card. It hid the very title the visitor wanted to click, and on a touch screen or with a keyboard there was no way to see it at all.

css/style.css

```
/* Now a visible thumbnail with a fixed proportion, so all cards
   measure the same even though the screenshots do not. */
.card-shot {
  width: 100%;
  aspect-ratio: 16 / 10;
  object-fit: cover;
  object-position: top center;
}

/* The whole card is clickable: this pseudo-element stretches the
   link's hit area over the card. It stays ONE link, so the keyboard
```

```
    and a screen reader see one stop, not a clickable box. */
.card { position: relative; }
.card-link::before {
  content: ""; position: absolute; inset: 0; z-index: 2;
}

.projects-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(290px, 1fr));
  gap: 1.25rem;
}
```

What it affects: with `auto-fit` the grid reflows on its own — four columns on a desktop, two on a tablet, one on a phone — with no media query at all.

My Courses

Category: 2026SEM02-UNICA - 10th semester

These are the three courses I am currently taking at Yachay Tech University during the second semester of 2026. For each one I list the instructor, the main topics of the syllabus, and the three things I personally expect to learn by the end of the term.

CA57 - Programación Web

COURSE CODE: CA57 · 2026SEM02UNICA-A00069

SEE IT IN THE SCHEDULE

WHERE THE CLASS IS

The course is built around real software development practices. We work with technical documentation and API references, framework documentation and database manuals, we manage our code in GitHub repositories, and we deploy the result to cloud platforms.

Main topics

- HTML & CSS
- JavaScript
- Frameworks
- Docker and virtual hosts
- Git and GitHub

Three topics I expect to learn

1. HTML, CSS, JavaScript, jQuery and Bootstrap
2. Postman for API testing
3. Educational platforms (LMS: Moodle)

Instructor

Prof. Francisco Javier Hidrobo Torres

External resource

Reference book I use for this course:
[Web Programming with HTML5, CSS and JavaScript - John Dean](#) (opens in a new tab).

CA56-TA24 - Procesos Estocásticos

COURSE CODE: CA56-TA24 · 2026SEM02UNICA-A00263

Figure 7. Courses: the page title is a band, each course a panel with a header painted in its own colour, and the sidebar carries what is consulted rather than read.

Two problems were fixed here. First, every section was the same white box, so the page title weighed the same as one of its courses. Second, the text column is capped at 68 characters to stay readable, but the section measured 1280px, so half the page was empty.

css/style.css

```
/* The page title is a band with a rule, not another box: the
   difference in FORM says which is which before anything is read. */
.section#intro {
  background: transparent; border: none;
  border-bottom: 3px solid var(--accent);
  padding: 0.5rem 0 1.75rem;
}

/* Widening the text was not an option: it reads worse. The freed
   space carries instructor, resource and links instead. */
.course-grid {
  display: grid;
  grid-template-columns: minmax(0, 40rem) minmax(0, 19rem);
  gap: 2rem;
  justify-content: start;
}
.course-side { position: sticky; top: 6rem; }
```

```
/* The subject colour moved from a 6px stripe to the whole header. */  
.c-web .course-head { background: var(--sub-web-bg); }  
.c-stoch .course-head { background: var(--sub-stoch-bg); }  
.c-phys .course-head { background: var(--sub-phys-bg); }
```

What it affects: the *sticky* sidebar stays in view while the syllabus scrolls past it, so the instructor and the reference book are readable at any point of a long course entry.

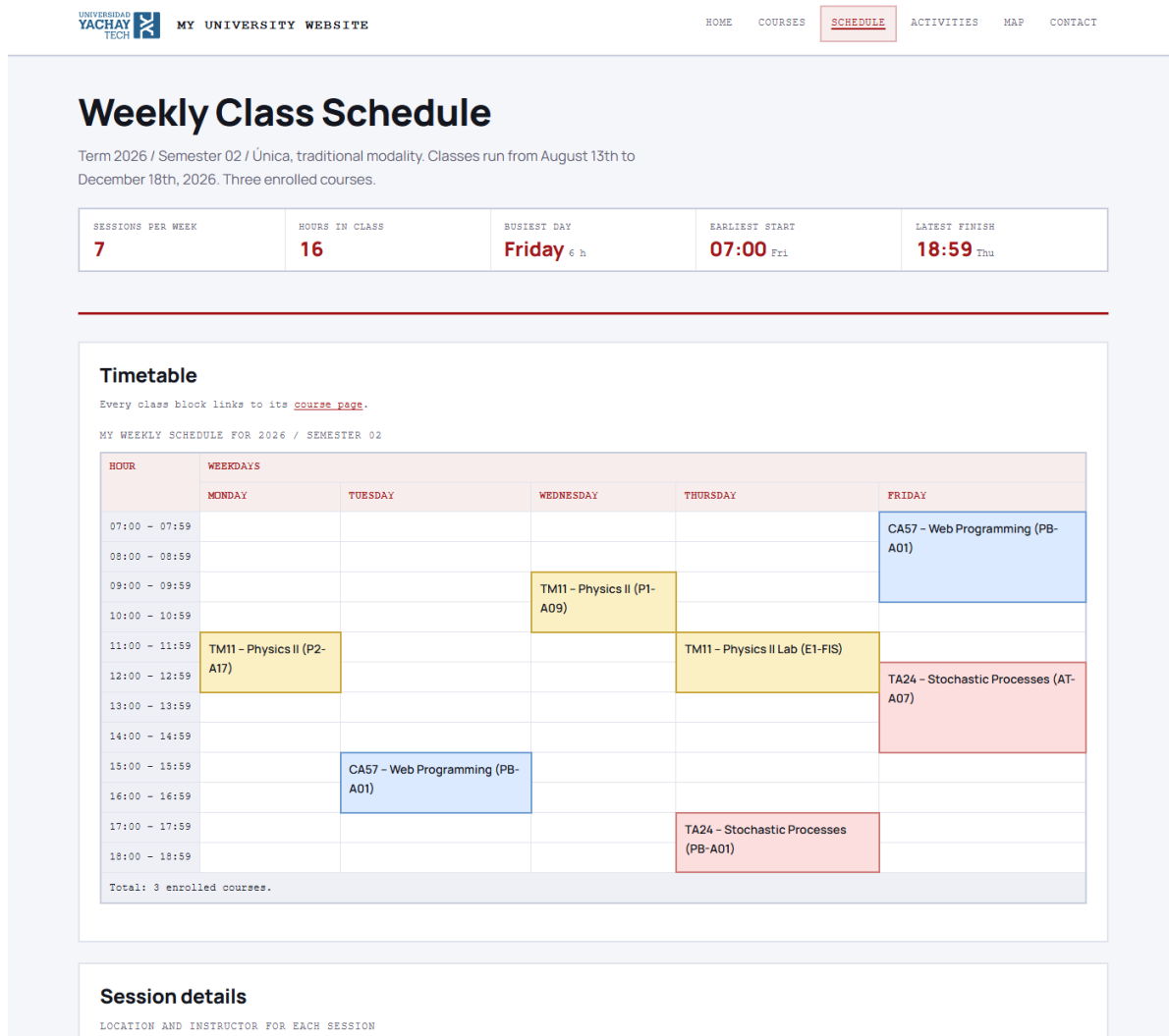


Figure 8. Schedule: five figures read off the timetable, and each class block in its subject colour.

Why the figures: the table answers “when is Physics”, but not “how heavy is my week”. Sessions, hours, busiest day, earliest start and latest finish were all in the table already — they just had to be counted by hand.

A colour per subject

css/style.css

```
/* Tokens, so the same colour reaches the timetable, the session
   table and the course page. */
:root {
  --sub-web-bg: #dbeafe; --sub-web-bd: #6b9bd1;
  --sub-stoch-bg: #fbf0c4; --sub-stoch-bd: #cc7b78;
  --sub-phys-bg: #fbf0c4; --sub-phys-bd: #c9a63a;
}

/* These selectors must match the weight of `.timetable td[rowspan]`
   (0,2,1). At lower weight the generic tint won and the whole
   timetable came out in the same red. */
.timetable td.c-phys, .sessions td.c-phys {
  background: var(--sub-phys-bg);
  border: 2px solid var(--sub-phys-bd);
}
```

}

Colour is never the only cue. Every cell still carries its course code written inside it, so someone who cannot tell the pastels apart loses nothing. In the dark theme the pastels are replaced by muted versions: a light pastel on a dark ground glares and the text on top stops being readable.

The generic rowspan tint is scoped to the timetable only. In the session table a rowspan merely groups rows that repeat a building or an instructor, and tinting those left half the table painted and half of it plain, depending on whether a value happened to repeat.

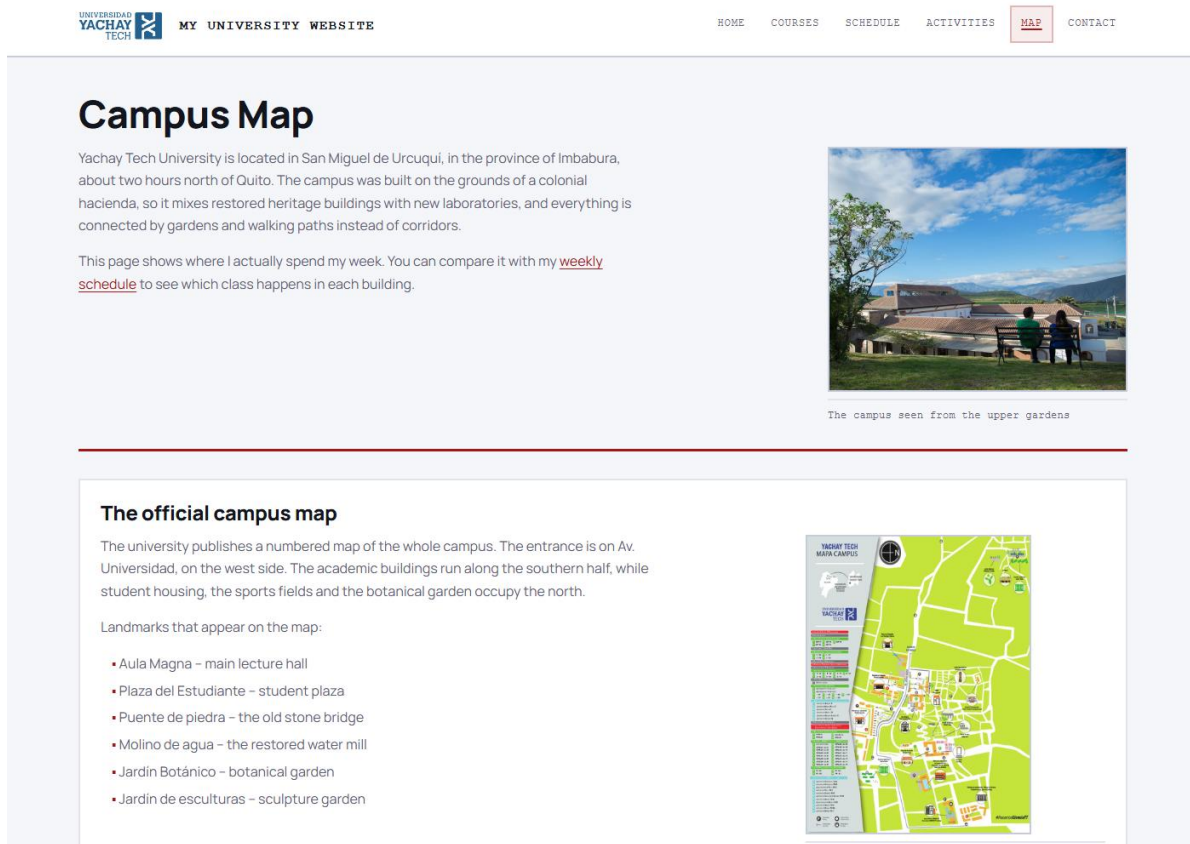


Figure 9. Map: text and figure side by side, with the campus photographs cropped to a shared proportion in the grids below.

Before: each figure took the full width under its paragraph. Several of these photographs are portrait (1000×1500), which at 520px wide is 780px of height per figure. The page grew downwards while the right half stayed empty.

css/style.css

```
.media-split {
  display: grid;
  grid-template-columns: minmax(0, 1fr) minmax(0, 22rem);
  gap: 2rem;
  align-items: start;
}

/* A height cap for standalone figures. `width: auto` keeps the
   proportion: the image shrinks, it does not distort. */
figure img { max-height: 26rem; width: auto; }

/* In the grids the thumbnails share a crop, so the rows line up even
   though the source photographs do not. */
.card-grid figure img {
  width: 100%; aspect-ratio: 4 / 3;
  object-fit: cover; max-height: none;
}
```

Result: the map page went from about 5000 pixels of height to 3364, with the same content.

Contact the University

This form lets a student write to Yachay Tech University: to student services, to the registrar's office, or to the academic department in charge of their programme. Fill in your academic details so that the request reaches the right desk.

Note: this form has no backend. Nothing is sent anywhere when you press the button. It exists as an exercise to learn how HTML forms and their input types work.

Student contact form

PERSONAL INFORMATION

Full name:
First and last name

Email address:
name@yachaytech.edu.ec

Date of birth:
dd/mm/aaaa

National ID number:
1000000000

Phone number (optional):
099 999 9999

ACADEMIC INFORMATION

Degree program:
- Select a program -

Semester:

REASON FOR CONTACT

☒ Academic records and certificates
☐ Enrollment and course registration
☐ Scholarships and financial aid
☐ Student housing
☐ Technical support
☐ Other

YOUR MESSAGE

Subject:

Message:
Describe your request in detail...

Attach a document (optional):
 Sin archivos seleccionados

Urgency:
Medium - within a few days

Figure 10. The form in two columns, with every label above its field.

Two problems. Five fieldsets stacked one under another read as a wall with no visible end. And the labels sat inline, so with names of different length ("Full name" against "Phone number (optional)") each field started at a different vertical.

css/style.css

```
.form-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(20rem, 1fr));
  gap: 0 1.5rem;
  align-items: start;
}

/* Block labels: one aligned column instead of a ragged edge. */
label { display: block; margin-bottom: 0.25rem; }

/* Radio and checkbox keep their label beside them, not above. */
input[type="radio"] + label,
input[type="checkbox"] + label { display: inline; margin-left: 0.35rem; }

input:focus, select:focus, textarea:focus {
  border-color: var(--accent);
}
```

```
box-shadow: 4px 4px 0 var(--accent);  
}
```

The form keeps the ten input types of Workshop 1 and still has no backend: nothing is submitted when the button is pressed.

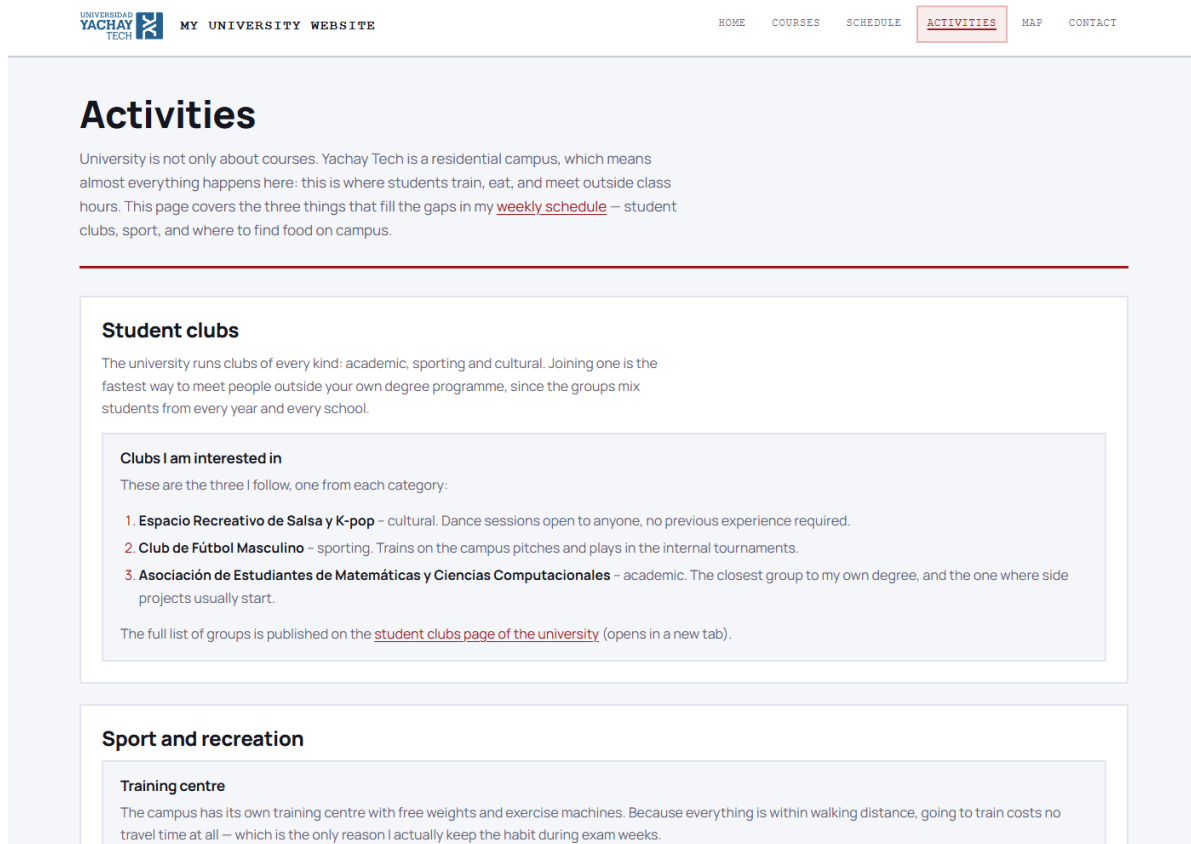


Figure 11. Activities: the articles laid out in a grid that reflows on its own.

Before: three lists one after another. It was the longest page with the least in it.

css/style.css

```
.card-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(19rem, 1fr));
  gap: 1.25rem;
  /* stretch, not start: with texts of different length the cards in a
    row ended at different heights and the bottom edge stepped. */
  align-items: stretch;
}
```

The image viewer every page with images

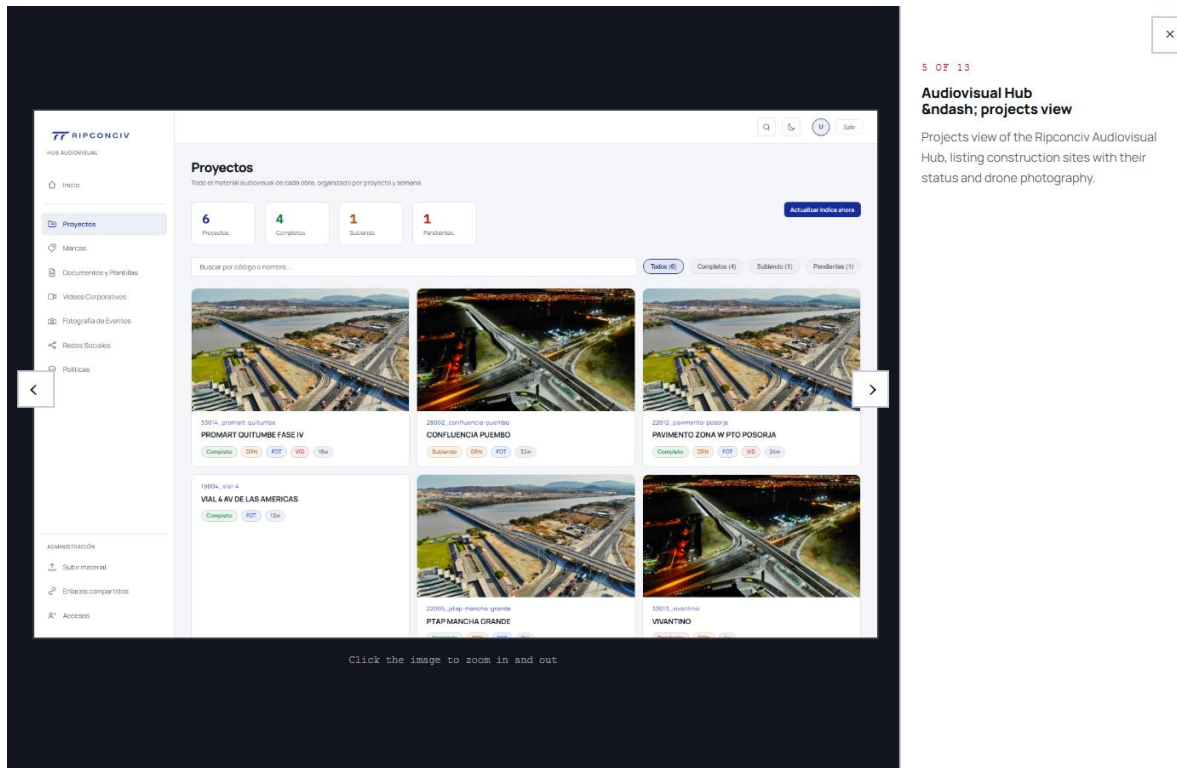


Figure 12. The viewer: the photograph on a dark stage, an information panel, a close button and arrows for the rest of the set.

Every content image opens full screen — 13 on the home page, 8 on the map. It has a close button, previous and next arrows, a panel with the caption and the description, and a zoom.

css/style.css

```
/* Same mechanism as the welcome screen: the viewer is a section that
   lives hidden until the URL points at it. */
.lightbox { display: none; }
.lightbox:target {
  display: grid;
  grid-template-columns: 1fr minmax(240px, 22rem);
  position: fixed; inset: 0; z-index: 100;
}

/* The zoom is a hidden checkbox and its label. It is the only way to
   have a switch in plain CSS: pressing the label checks the box, and
   `:checked ~` restyles the image. */
.lb-zoom-state:checked ~ .lb-frame img {
  max-width: none; max-height: none; /* natural size */
}
.lb-stage { overflow: auto; } /* so it can be panned */
```

A detail that mattered: the close button points at the anchor of the thumbnail it came from, not at href="#". With an empty fragment no :target is active, and the welcome screen came back every time a photo was closed.

4. Decisions that reach the whole site

Navigation between pages

The header was rebuilt. Logo, title and menu now share one centred container; before, the logo and the title sat outside the element that carried the width limit, so their left edges did not line up and the header took three rows.

css/style.css

```
.header-inner {
  max-width: var(--max-width); margin: 0 auto;
  display: flex; align-items: center; flex-wrap: wrap;
}
.nav { margin-left: auto; } /* menu to the far side */

/* The current page is marked in the HTML with aria-current="page", so
   a screen reader announces it and the CSS styles it from the same
   attribute: one source of truth, not two. The underline accompanies
   the colour, so the cue does not depend on colour alone. */
.nav-list a[aria-current="page"] {
  color: var(--accent);
  background: var(--accent-bg);
  text-decoration: underline;
}
```

Cross-page links were added where they were missing: every timetable block opens its course page, each course links to the schedule and to the map, and the map links back to both.

Micro-interactions

What makes a page feel alive is not decoration, it is that it responds. Each of these confirms a gesture.

css/style.css

```
/* The timetable row highlights across its full width: with twelve
   rows and five columns, losing your line is the easiest thing. */
.timetable tbody tr:hover th,
.timetable tbody tr:hover td:not([class]) { background: var(--bg3); }

/* Class blocks dim a touch: they lead somewhere. */
.timetable td[class]:hover { filter: brightness(0.94); }

/* Buttons sink by exactly the offset of their own shadow. */
.btn:hover { transform: translate(4px, 4px); box-shadow: none; }

/* The current menu entry deliberately does NOT move: you are already
   there, and moving it would suggest it leads elsewhere. */
```

Responsive

css/style.css

```
/* Grids reflow with auto-fit, so most of the site needs no
   breakpoint at all. One query handles the rest. */
@media (max-width: 640px) {
  body { font-size: 15px; }
  .nav { margin-left: 0; width: 100%; }
  .nav-list a { font-size: 0.7rem; }
  input, select, textarea { max-width: 100%; }
}

/* Wide content scrolls inside its own box, so the page body never
   scrolls sideways. */
```

```
.scroll-x { overflow-x: auto; }

/* 100dvh as well as 100vh: on phones the browser bar makes 100vh
   overshoot the screen. */
.intro { min-height: 100vh; min-height: 100dvh; }
```

Checked at a real 390-pixel viewport: the menu wraps, the tables scroll inside their own container, the grids reflow, and nothing overflows horizontally.

Accessibility

- **Skip link.** The first focusable element on every page jumps to the content, so a keyboard user does not tab through six menu entries on each page. It is moved off-screen rather than hidden with `display: none`, because a hidden element cannot take focus.
- **Touch targets.** Menu links and viewer controls are at least 44 pixels, the smallest comfortable size for a finger.
- **Visible focus.** Every interactive element shows a focus ring; none removes the outline without replacing it.
- **Reduced motion.** A `@media (prefers-reduced-motion: reduce)` block cancels the typing effect and every transition. The greeting still appears, fully typed.
- **Language.** Pages declare `lang="en"` and Spanish proper nouns are wrapped in `<span lang="es">`, so a screen reader switches pronunciation mid-sentence.

5. Verification

Every claim above was checked mechanically rather than by eye.

Check	Result
Tag balance, six pages	No unbalanced tags
Internal links	0 broken
Image references	0 broken
Stylesheet path from every page	Resolves
Text contrast, light theme	5.79:1 minimum (AA needs 4.5:1)
Text contrast, dark theme	5.74:1 minimum (AA needs 4.5:1)
Subject pastels, both themes	11.02:1 worst case
alt attributes	0 images without useful alt text
Heading hierarchy	One h1 per page, no skipped levels
label / for bindings	0 orphan labels
Layout at 390 px	No horizontal overflow
HTTP response, published site	200 on every page and asset

Contrast was computed from the token values with the WCAG relative luminance formula, compositing the semi-transparent table headers over their background first. Reading contrast off a screenshot is not reliable.

6. What would happen if we changed things

The stylesheet was built so that a few specific edits produce large, predictable changes. This is the difference between styling a page and building a system.

If the accent colour changed

Editing `--accent` in one place repaints the whole site. Nothing else needs touching, because no rule contains a literal colour.

If the stylesheet were removed

The site would look exactly like the Workshop 1 folder and would still be perfectly usable: every page reads top to bottom in order, the tables are still tables, the form still works. That is the reward for putting meaning in the HTML rather than appearance.

The only loss would be the welcome sequence and the viewer, which depend on `:target`. Even then nothing breaks: both would simply show as extra sections the visitor can scroll past.

If JavaScript were allowed

The welcome sequence and the viewer would be easier to write and easier to control. But the CSS-only version has a property the scripted one would lose: it works with scripting disabled, and it cannot break halfway. Every state is either in the URL or in an animation, and the browser handles both.

If the greeting had to be translated

The width of the typewriter is set to `12ch` because that is the length of “Hola, Mundo!”, and the animation runs in `steps(12)`. A translation of a different length needs both numbers changed together, or the text appears clipped or finishes early.

If more courses were added

A fourth subject needs three new tokens and one selector, following the pattern already there. What it must not do is rely on colour alone: the course code stays written in every cell.

Notes on the material used

- Campus photographs and the official campus map are material published by Yachay Tech University, reproduced here for an academic assignment.
- Screenshots of the Ripconciv and Centinela projects correspond to software developed by the author.
- No JavaScript is used anywhere on the site.